<table>
<tr><th colspan="2">BUSINESS CASE</th></tr>
<tr><td>Proposed Project</td><td>Ecosystem Discovery Application

A web-based educational application designed for elementary school children that introduces freshwater animals through simple, interactive discovery, with an emphasis on basic ecosystem understanding and conservation awareness.</td></tr>
<tr><td>Date Produced</td><td>27 January 2026</td></tr>
<tr><td>Background</td><td>Environmental education for elementary school children is often delivered through textbooks or static digital content, which can limit engagement and make ecosystem concepts difficult to understand. Freshwater ecosystems are among the most familiar and relevant environments for young learners, yet there are few age-appropriate, interactive tools that allow children to explore freshwater animals in a simple and meaningful way.

This project proposes an educational web application that introduces freshwater ecosystems through interactive discovery, supporting early environmental awareness and curiosity-driven learning.</td></tr>
<tr><td>Business Need/ Opportunity</td><td>There is a need for engaging age-appropriate educational tools that help elementary school children actively learn about freshwater ecosystems and conservation. Traditional approaches rely heavily on passive learning, which can reduce understanding and retention.

This project presents an opportunity to enhance early environmental education by providing an interactive, web-based application that encourages exploration and supports foundational ecosystem learning.</td></tr>
<tr><td>Options</td><td>Option 1: Develop the Ecosystem Educational Discovery Application
Design and implement a web-based educational application
Focus exclusively on freshwater animals
Tailor content and interactions for elementary school children
Follow an MVC architecture with a database-backed model
Deliver a clearly defined Minimum Viable Product (MVP)
Option 2: Do Nothing
Continue relying on existing static educational resources
No interactive or discovery-based learning experience
No improvement in student engagement or learning outcomes</td></tr>
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Cost-Benefit Analysis

Option 1: Develop the Application
Costs:

- Time and effort invested by team members
- Learning and coordination required for design and development
- Potential technical challenges during implementation

Benefits:

- Increased engagement for young learners
- Improved understanding of freshwater ecosystems
- Supports interactive and exploratory learning
- Reinforces conservation awareness at an early age
- Meets course learning objectives related to software engineering, MVC architecture, and project management
- Produces a portfolio-quality academic project

Option 2: Do Nothing

Costs:

- No development or implementation effort required

Benefits:

- No short-term resource investment
- No educational improvement or innovation is achieved

Recommendation

It is recommended to proceed with Option 1: Development of the Ecosystem Educational Discovery Application.
This option provides meaningful educational value by introducing elementary school children to freshwater animals through interactive learning. The project is appropriately scoped for an academic timeline, aligns with course requirements, and offers both educational benefits for users and practical learning outcomes for the project team.